



Spendly

Offline-First Zero-Knowledge Peer-to-Peer Finance Ecosystem

[Student Name] · [Enrollment Number] · Semester VII · Information Technology · [College Name]



THE PROBLEM

Finance Is Broken by Design

Privacy at Risk

Every transaction, wallet balance, and spending habit is uploaded to centralized servers — exposed to breaches, surveillance, and misuse.

Cloud Dependency

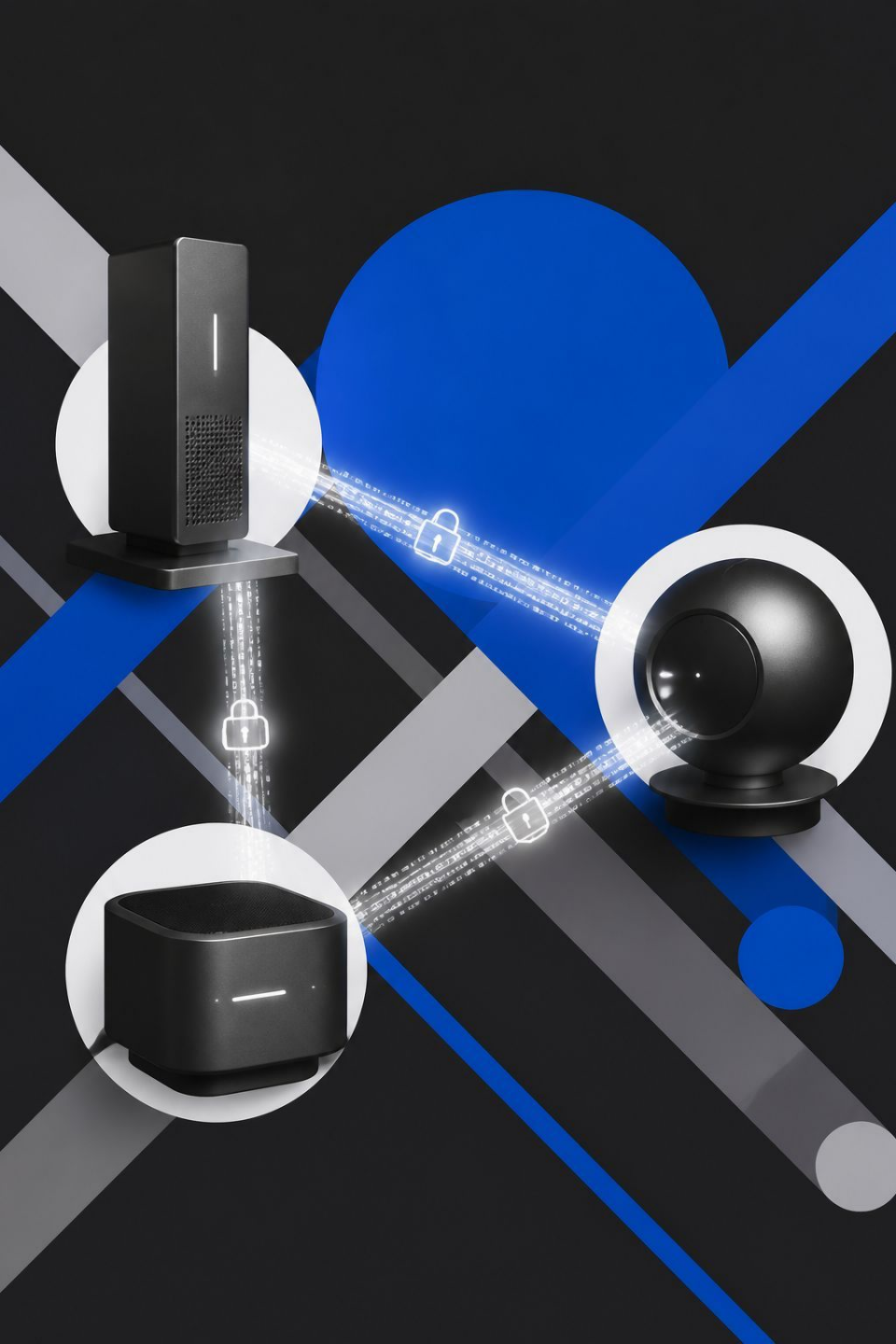
Internet outages halt billing, inventory, and payments. Businesses lose revenue every time connectivity drops.

Centralized Databases

A single point of failure. One breach exposes millions of users' financial identities simultaneously.

Expensive POS Systems

Traditional Point-of-Sale systems demand monthly subscriptions, hardware costs, and internet connectivity — locking small merchants out.



THE SOLUTION

Three Apps. One Ecosystem. Zero Compromise.

Spendly

Personal Finance PWA — expense tracking, wallet management, EMI reminders, and complete financial privacy. Data stays on-device.

Spendly Shop

Offline Merchant POS — scan products, generate GST invoices, apply discounts, and deliver receipts via WebRTC, QR, or print.

Spendly Radar

Merchant Management Hub — inventory control, stock updates, offer management, and anonymous local business analytics.

System Architecture

Frontend Layer

React · Vite · Tailwind CSS · Zustand · Framer Motion

Offline Storage

IndexedDB · Dexie.js — encrypted local-first database

Peer Network

WebRTC · PeerJS · ntfy.sh — serverless device-to-device communication

Edge & Security

Cloudflare Workers · Worker KV · WebAuthn · AES-256-GCM

Every layer operates independently. No single point of failure. No centralized data store. Complete privacy by architecture.



ECOSYSTEM

How the Spendly Ecosystem Works

Spendly → Personal Finance

Records expenses, manages wallets, stores invoices locally. Receives encrypted invoices from Spendly Shop via WebRTC. All data protected by WebAuthn biometric authentication.

Spendly Shop → Merchant Billing

Scans products, calculates GST, generates professional invoices. Retrieves pricing from Spendly Radar via peer-to-peer. Delivers receipts via WebRTC, QR, WhatsApp, or print.

Spendly Radar → Management Hub

Stores products, inventory, and campaigns locally. Responds instantly to Shop requests. Updates stock post-checkout. Generates anonymous business insights — zero customer data stored.



TECHNOLOGY

Technology Stack

Frontend

React · Vite · Tailwind CSS · Zustand ·
Framer Motion

Backend

Cloudflare Workers · Worker KV —
serverless edge compute

Networking

WebRTC · PeerJS · ntfy.sh — peer-
to-peer without servers

Security

WebAuthn · AES-256-GCM — biometric + military-grade
encryption

Database

IndexedDB · Dexie.js — offline-first local storage

SECURITY

Zero Knowledge. Zero Compromise.



Zero-Knowledge Architecture

No financial data ever leaves the device. The server knows nothing about your transactions.



Offline-First Design

Full functionality without internet. Data syncs peer-to-peer when devices are in range.



WebAuthn Biometric Unlock

Hardware-backed authentication. Face ID, Touch ID, or Windows Hello protect every session.



AES-256-GCM Encryption

Military-grade encryption for all stored and transmitted financial data.





FEATURES

Everything You Need. Nothing You Don't.



Expense Tracking & Analytics

Real-time dashboards, wallet balances, and spending insights — generated locally, never uploaded.



Merchant Billing & QR Invoices

Scan, calculate GST, apply discounts, and deliver invoices via WebRTC, QR Code, or WhatsApp.



Inventory & Stock Management

Track products, update stock post-checkout, and manage promotional campaigns from Radar.



Cash Wallet & EMI Reminders

Manage cash balances and track EMI schedules — all stored securely inside IndexedDB.



WHAT'S NEXT

Advantages & Future Scope

Today's Advantages

- **Offline-First** — fully functional without internet
- **Zero Server Cost** — no infrastructure, no subscriptions
- **Privacy by Design** — zero-knowledge, on-device storage
- **Fast Billing** — instant peer-to-peer invoice delivery
- **Secure Communication** — encrypted WebRTC channels

Future Roadmap

- **AI Analytics** — intelligent spending insights
- **Voice Billing** — hands-free merchant interaction
- **NFC & UPI** — contactless payment integration
- **GST Compliance** — automated tax filing
- **Multi-Store** — chain merchant support

Spendly

Privacy-First. Offline-Ready. Peer-to-Peer.

Spendly redefines personal and merchant finance through zero-knowledge architecture, offline-first design, and encrypted peer-to-peer communication — no centralized servers, no data exposure, no compromise.

Thank You

Questions & Answers